Qindong Gan

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EDUCATION

Waseda University **Tokyo, Japan**

Master of Engineering in Computer Science and Communications Engineering *09/2025-08/2027 (Expected)*

- **GPA: 3.33/4.00 (Waseda scale; A+ in 8 of 12 completed courses)**

University of Cincinnati **Cincinnati, OH, US**

Bachelor of Science in Electrical Engineering *09/2020-05/2025*

- **GPA: 3.571/4.000**

Joint program with Chongqing University; final academic year in Cincinnati.

Chongqing University **Chongqing, China**

Bachelor of Engineering in Electrical Engineering and Automation *09/2020-06/2025*

- **GPA: 3.35/4.00**

PUBLICATIONS

- **Q. Gan**, "When Self-Reported Fitness Becomes the Target: Independent Audits in LLM-Driven Recipe Evolution." **Submitted to** *Swarm and Evolutionary Computation* (SWEVO; JCR Q1). **Under review.**
- **Q. Gan**, Z. Pan, T. D. Ho, J. Wu, and S. Shimamoto, "Testing Feedback in QoS Pricing for Decentralized SAGIN Access Association," *IEEE Consumer Communications & Networking Conference (CCNC)*, 2027. **Accepted.**

PROJECTS

Testing Feedback in QoS Pricing for Decentralized SAGIN Access Association [Link] **03/2026-08/2026**

First-author research (IEEE CCNC 2027) | Advisor: Shigeru Shimamoto, Waseda University

- **Problem:** Addressed intermittent service starvation in decentralized SAGIN access association, where vehicles select LEO satellite, HAPS, or terrestrial links under co-tier interference. Long-term average-rate constraints can mask severe short-term service shortfalls.
- **Method:** Developed a QoS pricing approach incorporating per-vehicle expected-shortfall budgets into MAPPO-style centralized training and decentralized execution. Used one-sided shortfall costs to prevent surplus service from canceling deficits, with a consistent budget residual in both training rewards and Lagrangian price updates.
- **Experiments:** Evaluated 27 policies trained for 100,000 environment steps each across three training seeds. Used unpriced, residual-shuffling, and matched-initialization adaptive/frozen-price controls to examine service improvements and price-feedback mechanisms, with three budget settings to assess sensitivity.
- **Results:** At 12 vehicles and a budget of 0.377, reduced mean outage from **31.42% to 6.88%** and increased the mean instantaneous minimum rate from **0.0116 to 0.2097 Mbps** relative to unconstrained MAPPO, averaged over three training seeds. These service gains involved lower aggregate throughput and more frequent association changes. Matched controls showed that residual shuffling retained the gains and that benefits of price adaptation depended on the budget.

Keywords: Constrained MARL; MAPPO; Lagrangian dual ascent; price-feedback evaluation; QoS fairness; SAGIN.

ShinsaEvolve: Independent Audits in LLM-Driven Recipe Evolution [Link] **09/2026-10/2026**

Independent research project; sole-author manuscript submitted to SWEVO

- **Problem:** LLM-driven program evolution selects candidates by self-reported scores; when each RL training recipe controls its own reward shaping and evaluation schedule, those scores lose a common scale and selection can reward the report rather than the policy.
- **Innovation:** Designed a harness-owned audit that scores only the trained policy parameters on 32 fresh seeds with raw environment reward, never executing candidate code; released an open, reproducible system with immutable run configurations, prompt hashing, and per-episode audit records.
- **Experiments:** Ran a 2×2 LunarLander study (audit vs. self-reported selection $\times$ candidate-shaped vs. fixed raw reward) with 12 outer-loop runs and 180 LLM-generated candidates trained by CMA-ES.
- **Results:** Under candidate-shaped reward, self-reported scores saturated and the selected policy averaged -762.1 audited return vs. 4.0 under audit selection; fixed raw reward narrowed the gap (-137.1 vs. -19.4) but misselection persisted in all three runs.

Keywords: LLM-driven program evolution; reinforcement learning; evaluation reliability; reward hacking; evolution strategies.

RGB-Thermal Semantic Segmentation for Autonomous Driving [Link] **07/2025-08/2025**

Full-time AI Algorithm Intern | Supervised by Jiachen Liang, SERES Group Co., Ltd.

- **Goal:** Pixel-level segmentation of driving scenes (e.g., road, vehicles, pedestrians) by fusing visible and thermal infrared images.
- **Approach:** Adapted DFormerV2's geometry-prior design, originally developed for RGB-D segmentation, to RGB-thermal input.
- **Work:** Contributed to project data collection; built, trained, and validated the RGB-thermal segmentation model.

Keywords: Multimodal semantic segmentation; RGB-thermal fusion; autonomous driving; DFormerV2.

RoboNeuro Mapping: Bio-inspired Visual Navigation for Robots [Link] **09/2024-05/2025**

Senior Capstone Project (team of three) | University of Cincinnati | Advisor: Ali Minai

- **Goal:** Extend a self-organizing hippocampal place-cell navigation model, previously driven by LiDAR alone, with camera input so that visual landmarks can disambiguate similar spatial contexts (Webots; iRobot Create 3).
- **Approach:** Embedded pretrained EfficientNetV2B0 visual features into the place-cell state space alongside LiDAR-based spatial representations for goal-directed navigation.
- **Work:** Collected and preprocessed simulation images, validated the image pipeline, and contributed to visual feature extraction and evaluation; coordinated team discussions and weekly progress reporting.

Keywords: Bio-inspired navigation; hippocampal place cells; visual–LiDAR fusion; EfficientNetV2; Webots.

Data-Driven Condition Monitoring of Power Electronics [Link] **05/2024-07/2024**

Research Assistant | Chongqing University | Advisor: Wei Lai

- **Goal:** Estimate capacitor aging condition robustly under measurement noise, sensor error, and changing operating conditions.
- **Work:** Ported the group's MATLAB sliding-window, wavelet time-frequency, and CNN-regression pipeline to Python with TensorFlow/Keras; integrated the model into LabVIEW via Python Node and tested it in simulation under noise and sensor errors.

Keywords: Condition monitoring; power electronics; time-frequency features; CNN regression; LabVIEW.

RESEARCH INTERESTS

Multi-agent reinforcement learning; decentralized decision-making; reliable policy evaluation.

HONORS AND AWARDS

- Best Master's Student Presentation Award, ASPIRE-CRC-TCA International Workshop, Memorial University, 2026.
- Dean's List, University of Cincinnati, Fall 2024 and Spring 2025.
- Cooperative Education & Practice - Excellence, University of Cincinnati.
- Second Prize, National College Student Robot Competition.
- Outstanding Student Scholarship (Top 10%); Outstanding Graduate, Chongqing University.

PRESENTATIONS AND TEACHING EXPERIENCE

Workshop Presentation **08/2026**

- "Multi-Agent Reinforcement Learning for Intelligent Access Management in Space-Air-Ground Integrated Networks." ASPIRE-CRC-TCA International Workshop, Memorial University.

Teaching Assistant, University of Cincinnati **09/2024-05/2025**

Courses: Foundations of Engineering Design Thinking I & II (ENED 1100 / ENED 1120).

- Supported undergraduate engineering instruction in English through recitations and labs; graded assignments and examinations.

SKILLS

Programming and tools: Python, MATLAB, PyTorch, TensorFlow/Keras, LabVIEW, EPyMARL, NumPy, pandas, scikit-learn, Git, LaTeX.

Methods: Multi-agent and constrained reinforcement learning, Lagrangian methods, simulation, ablation studies, multimodal perception.

Languages: Mandarin Chinese (native), English.